Name – Pranav Kumar Verma

Roll – 2401331540186

Branch – DS(C)

Q4) Train an ANN with different learning rates and compare the convergence of the model

Ans :-

```
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Flatten, Dense
from tensorflow.keras.optimizers import Adam
import matplotlib.pyplot as plt
(X_train, y_train), (X_test, y_test) = tf.keras.datasets.mnist.load_data()
X_train = X_train.astype("float32") / 255
X_test = X_test.astype("float32") / 255
learning_rates = [0.1, 0.01, 0.001]
histories = []
for lr in learning_rates:
    print(f"\nTraining Model with Learning Rate = {lr}")

    model = Sequential([
        Flatten(input_shape=(28, 28)),
        Dense(128, activation='relu'),
        Dense(64, activation='relu'),
        Dense(10, activation='softmax')
    ])

    optimizer = Adam(learning_rate=lr)

    model.compile(
        optimizer=optimizer,
        loss='sparse_categorical_crossentropy',
        metrics=['accuracy']
    )

    history = model.fit(
        X_train,
        y_train,
        epochs=5,
        batch_size=32,
        validation_split=0.2,
        verbose=1
    )
```

```
        loss, accuracy = model.evaluate(X_test, y_test, verbose=0)

        print(f"Learning Rate: {lr}")
        print(f"Test Accuracy: {accuracy:.4f}")

        histories.append(history)
    plt.figure(figsize=(8,5))

    for i, lr in enumerate(learning_rates):
        plt.plot(histories[i].history['loss'], label=f'LR={lr}')

    plt.title("Training Loss vs Epochs")
    plt.xlabel("Epoch")
    plt.ylabel("Loss")
    plt.legend()
    plt.grid(True)
    plt.show()
    plt.figure(figsize=(8,5))

    for i, lr in enumerate(learning_rates):
        plt.plot(histories[i].history['val_accuracy'], label=f'LR={lr}')

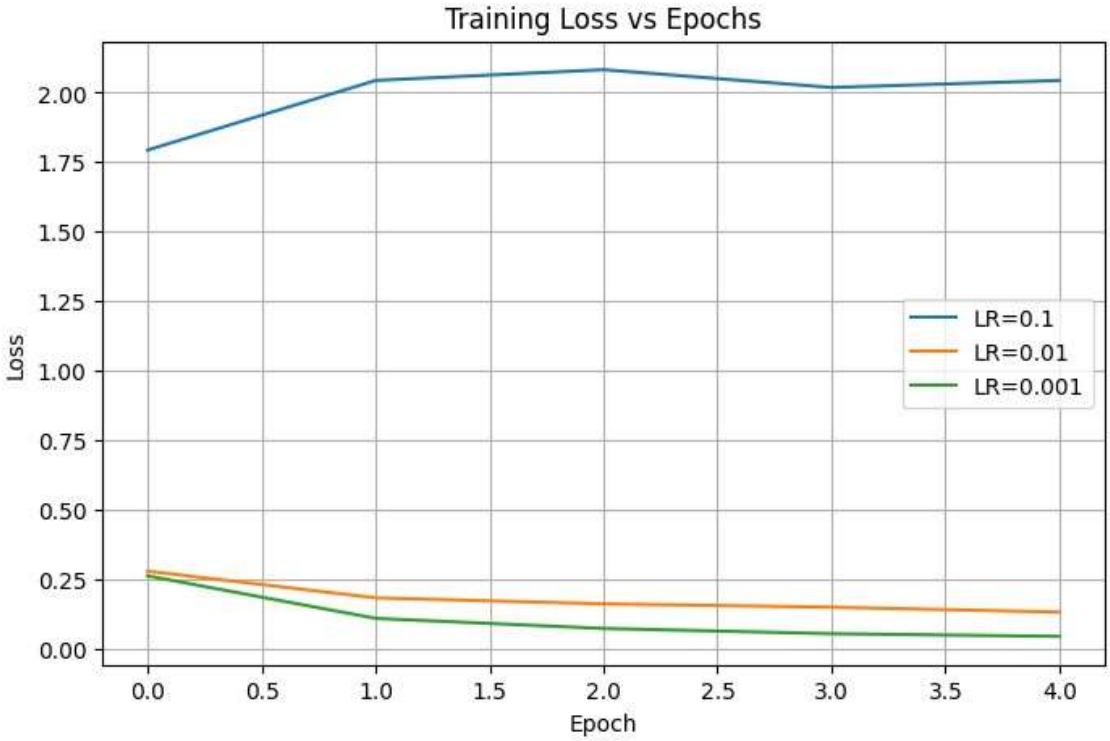
    plt.title("Validation Accuracy vs Epochs")
    plt.xlabel("Epoch")
    plt.ylabel("Accuracy")
    plt.legend()
    plt.grid(True)
    plt.show()
```

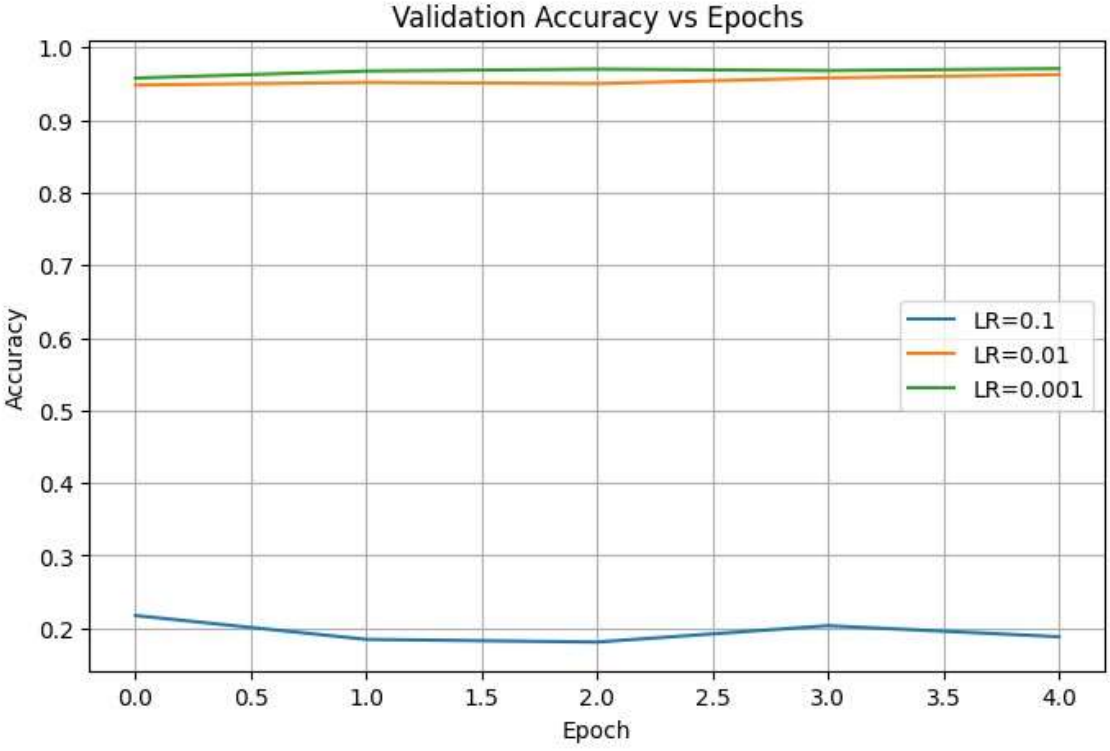
```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step

Training Model with Learning Rate = 0.1
/usr/local/lib/python3.12/dist-packages/keras/src/layers/reshaping/flatten.py:37: UserWarning: Do not pass an `input_shape`/`input_dim` argumen
  super().__init__(**kwargs)
Epoch 1/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 10s 6ms/step - accuracy: 0.3346 - loss: 1.7939 - val_accuracy: 0.2174 - val_loss: 1.9927
Epoch 2/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 7s 5ms/step - accuracy: 0.1996 - loss: 2.0440 - val_accuracy: 0.1846 - val_loss: 2.0271
Epoch 3/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 9s 6ms/step - accuracy: 0.1858 - loss: 2.0827 - val_accuracy: 0.1809 - val_loss: 2.0693
Epoch 4/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 9s 6ms/step - accuracy: 0.1975 - loss: 2.0187 - val_accuracy: 0.2034 - val_loss: 2.0168
Epoch 5/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 7s 5ms/step - accuracy: 0.1964 - loss: 2.0439 - val_accuracy: 0.1883 - val_loss: 2.0571
Learning Rate: 0.1
Test Accuracy: 0.1972

Training Model with Learning Rate = 0.01
Epoch 1/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 10s 6ms/step - accuracy: 0.9176 - loss: 0.2794 - val_accuracy: 0.9483 - val_loss: 0.1743
Epoch 2/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 8s 6ms/step - accuracy: 0.9495 - loss: 0.1835 - val_accuracy: 0.9519 - val_loss: 0.1916
Epoch 3/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 8s 5ms/step - accuracy: 0.9561 - loss: 0.1620 - val_accuracy: 0.9503 - val_loss: 0.2194
Epoch 4/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 9s 6ms/step - accuracy: 0.9603 - loss: 0.1500 - val_accuracy: 0.9582 - val_loss: 0.1526
Epoch 5/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 7s 5ms/step - accuracy: 0.9653 - loss: 0.1324 - val_accuracy: 0.9628 - val_loss: 0.1554
Learning Rate: 0.01
Test Accuracy: 0.9625

Training Model with Learning Rate = 0.001
Epoch 1/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 9s 5ms/step - accuracy: 0.9229 - loss: 0.2625 - val_accuracy: 0.9579 - val_loss: 0.1426
Epoch 2/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 9s 6ms/step - accuracy: 0.9664 - loss: 0.1097 - val_accuracy: 0.9676 - val_loss: 0.1081
Epoch 3/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 9s 6ms/step - accuracy: 0.9775 - loss: 0.0735 - val_accuracy: 0.9703 - val_loss: 0.1003
Epoch 4/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 8s 5ms/step - accuracy: 0.9830 - loss: 0.0549 - val_accuracy: 0.9682 - val_loss: 0.1061
Epoch 5/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 10s 6ms/step - accuracy: 0.9858 - loss: 0.0450 - val_accuracy: 0.9709 - val_loss: 0.0978
Learning Rate: 0.001
Test Accuracy: 0.9734
```


Training Loss vs Epochs
Loss
Epoch
LR=0.1
LR=0.01
LR=0.001
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0


Validation Accuracy vs Epochs
Accuracy
Epoch
LR=0.1
LR=0.01
LR=0.001
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0